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The impact of methane inhibitors on ruminants: A systematic review and meta-analysis

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ABSTRACT

The impact of methane inhibitors on ruminant performance and rumen microbial community composition is unclear. The aim of this study was to summarize the effects of methane inhibitors on the performance of ruminants and the structure of rumen microbial communities. A total of 13,043 studies were retrieved from the Web of Science database. Ultimately, 256 studies containing the variables we needed were included. The data were further extracted and processed. The study revealed the negative effects of methane inhibitors on ruminants, which were reflected in the reduction of feed intake and digestibility in ruminants. Adding methane inhibitors reduced the acetate concentration in the rumen and increased the propionate content in the rumen. There was no significant change in the α-diversity of the rumen microbiome, whereas the β-diversity of rumen microbes was enhanced. The effects of methane inhibitor supplementation showed dose-dependent significant differences, particularly in modulating rumen fermentation parameters and the structure of the microbial community. Furthermore, when the total VFA in the rumen were below 96.98 mmol/L, or the acetate concentration was below 61.26 mmol/L, or the acetate-to-propionate ratio was below 3.86, the suppression of methane production was most effective. The addition of methane inhibitors has a positive effect on the performance of ruminants, particularly by improving the structure of the rumen microbiota. Additionally, VFA have a certain threshold effect on methane production. This provides a reference for the application and selection of methane inhibitors.

Key words: ruminant, methane inhibitor, rumen microbiota, meta-analysis

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INTRODUCTION

The methane emissions from ruminants account for about one-third of global anthropogenic GHG emissions, with 88% originating from the digestive tract of ruminants and the remainder from manure fermentation, and the energy loss resulting from this process accounts for 2% to 12% of the total energy intake of ruminants (Ungerfeld and Pitta, 2025; Xie et al., 2025). This will not only lead to a further increase in global temperatures but also result in a decrease in feed energy utilization efficiency. The application of methane inhibitors is expected to alleviate this dilemma. The current methane reduction strategies mainly include pasture management, diet regulation, and the addition of methane inhibitors (Smith et al., 2022). The main types of methane inhibitors include lipids, ionophores, nitrates, plant secondary metabolites, saponins, essential oils, tannins, halogenated methane analogs, probiotics, prebiotics, and others (Fouts et al., 2022).

The production of methane in the rumen occurs in 3 main stages. In the first stage, after ruminants consume their diet, hydrolytic or fermentative microbial communities break down the polymers in the cell walls into organic acids, alcohols, hydrogen, and carbon dioxide. In the second stage, acetate-producing microbes use the products from the first stage to generate acetate and hydrogen, providing energy for bacterial growth. In the third stage, methane-producing archaea use the products from the previous 2 stages to generate methane and carbon dioxide (Mackie et al., 2024). The electrons produced during rumen fermentation combine with NAD^+, and these electrons are transferred by microorganisms carrying H_2-evolving hydrogenases in the rumen. They couple with reduced ferredoxin to produce H_2 (Xie et al., 2025). The pyruvate produced from carbohydrate metabolism in the rumen tends to form acetate when the partial pressure of hydrogen is low. This is because the thermodynamic reaction of NADH (reduced form) oxidation to produce hydrogen gas is nonspontaneous, unless the hydrogen concentration is low, in which

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

case the hydrogen gas produced by this reaction is used by methanogens to produce methane. When the hydrogen concentration is high, pyruvate tends to form propionate and even butyrate, iso-butyrate, lactic acid, and other products. The hydrogenotrophic methanogens generate methane using hydrogen and methyl or carbon dioxide under the action of methyl-coenzyme M reductase (MCR; Pitta et al., 2022a). The hydrogenotrophic pathway is the main route for methane production in the rumen (Mackie et al., 2024). The MCR of methanogenic archaea is composed of 3 subunits: α, β, and γ, which are encoded by the *mcrA*, *mcrB*, and *mcrG* genes, respectively. In the final step of methane generation, it reduces substrates such as carbon dioxide, formate, methyl compounds, and acetate to methane. The inhibition of MCR activity is considered a key point in controlling methane emissions (Kumar et al., 2009; Duin et al., 2016).

According to current research, methods for reducing methane emissions can be divided into 2 categories. The first category is to reduce the availability of hydrogen (Hristov et al., 2022). When the rumen tends to produce propionate, protons are consumed, whereas the production of acetate is a process that generates protons (Czatzkowska et al., 2020). Therefore, propionate-type fermentation leads to a reduction in methane production. For example, a high-concentrate diet can lead to an increase in hydrogen partial pressure, enhanced pyruvate metabolism, increased production of reductive propionate, and a decrease in the concentrations of acetate, carbon dioxide, and other compounds (Mackie et al., 2024). Additionally, providing hydrogen acceptors or sinks is also an important method for reducing methane production (Sejian et al., 2011). Representative substances are nitrates, which are competitive electron acceptors with methyl groups and carbon dioxide, offsetting the reducing equivalents of hydrogen during fermentation (Božic et al., 2009). Another important method is to provide inhibitors that directly suppress the activity of methane-producing microorganisms in the rumen, thereby inhibiting methane production. For example, medium- and long-chain fatty acids inhibit methane production by disrupting the cell membranes of methanogens, thereby reducing their activity. Among them, medium-chain fatty acid lauric acid inhibits methane production by 89% in vitro and by 76% in vivo (Machmüller et al., 2002; Soliva et al., 2003; Božic et al., 2009). Furthermore, MCR, as the key enzyme catalyzing the final step of methane production by reducing methyl-CoM to methane, plays a central role in the methane generation pathway. Due to its significance, many studies target MCR as a primary point of inhibition to reduce methane production (Dinh and Allen, 2024).

According to current research, there is no consensus on the effect of methane inhibitors on ruminants. The effects of methane inhibitors on ruminants cannot be generalized. This is mainly influenced by various factors such as the type of methane inhibitor, the dosage of the inhibitor, the species of livestock, and the physiological condition of the animals (Liu et al., 2025). Furthermore, even for the same type of inhibitor, contradictory research results may exist. For example, in Almeida et al. (2023), the final BW increased gradually with the addition of 3-nitrooxypropanol (**3-NOP**), although the difference was not significant, whereas the results from Meale et al. (2021) were opposite this result. There are also some common issues regarding the effect of the same type of inhibitor on the rumen microbial structure. For example, in Patra and Yu (2015), saponins led to a significant increase in the total number of bacteria in the rumen, whereas in Ramírez-Restrepo et al. (2016), a notable significant decrease in the total bacterial count was observed. Therefore, the aim of this study is to collect data from published research and conduct a comprehensive analysis of the effects of methane inhibitors on ruminant intake, digestion, growth, rumen fermentation, and the rumen microbial community structure.

MATERIALS AND METHODS

Data Collection

To streamline the process and minimize errors associated with screening duplicate records across multiple databases, the literature search was confined to the Web of Science database (https://www.webofknowledge.com/). All searches were completed on February 18, 2025. The search keywords include all ruminant livestock, and free word searches were conducted on the MeSH module (https://www.ncbi.nlm.nih.gov/mesh/) of the NCBI database (https://www.ncbi.nlm.nih.gov/). Given the potential selection bias arising from exclusively using the Web of Science for literature screening, we implemented a broader search strategy. The subject heading, free text, and search strategy are provided in Supplemental Table S1 (see Notes). The scope of the search needs to be specified as "Topic." For the studies that satisfy the inclusion criteria, extract the mean values of the outcomes for both the experimental and control groups. The included references can be found in Supplemental Table S2 (see Notes). Data presented in graphical form were extracted using the software tool GetData Graph Digitizer v.2.24 (GetData Pty Ltd.). For boxplots, the sample mean was estimated based on the quantile estimation of the boxplot (McGrath et al., 2020).

Study Eligibility

The study inclusion process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses. The flowchart of the search and exclusion process can be found in Supplemental Figure S2 (see Notes). The inclusion criteria were strict: (1) the study must use only one methane inhibitor as an additive; (2) both a control group and an experimental group must be included; (3) the pretreatment for both the control and experimental groups must be consistent; (4) the treatment or additive must significantly reduce methane emissions; and (5) the data must be extractable from tables or figures. Two authors conducted an independent review of the retrieved articles and engaged in discussions regarding any discrepancies. Only when a consensus was reached between the 2 reviewers would the study be included. In conclusion, a total of 256 studies were retained, and the location details of the studies are provided in Supplemental Figure S2. To test for publication bias in the included literature, a funnel plot of the effect sizes was drawn, and Egger's test was performed using R software version 4.2.3, which can be found in Supplemental Figure S1 (see Notes).

Data Analysis

The rumen microbiota data collected from the first 2 data axes of the ordination analysis needed further processing. The Euclidean distances within the control group, within the experimental group, and between the control and experimental groups were calculated. Additionally, the β-diversity and community structure of the rumen microbiota were further calculated using the following formula (Li et al., 2024):

$$\mathrm{RR}_{Structure} = \overline{D_b} \Big/ \left(\overline{D_c + D_t}\right),$$

$$\mathrm{RR}_{\beta} = \overline{D_b} \Big/ \overline{D_c},$$

where RR represents the response ratios, D_c represents the Euclidean distance within the control group, D_t represents the Euclidean distance within the experimental group, and D_b represents the Euclidean distance between the control and experimental groups. If $\mathrm{RR}_{structure} < 0$, it indicates that the methane inhibitors have no effect on the rumen microbial community structure, with higher positive values representing a greater effect. Similarly, when $\mathrm{RR}_{\beta} < 0$, it indicates a decrease in rumen microbial β-diversity, whereas the values >0 indicate an increase in rumen microbial β-diversity.

Due to the inconsistent units of the variable across all the collected studies, a natural logarithm transformation was applied for standardization (Zhou et al., 2020). When the transformed value is greater than zero, it indicates a positive effect of the treatment

$$\mathrm{RR} = \ln\left(\overline{X_t} \Big/ \overline{X_c}\right),$$

where $\overline{X_t}$ and $\overline{X_c}$ represent the mean values of a certain variable under the methane inhibitor treatment group and the control group, respectively. The calculation formula for the weighted value of each variable was

$$W = (n_t \times n_c)/(n_t + n_c),$$

where W is the weighted value of the variable, and n_t and n_c represent the number of replicates for the methane inhibitor treatment group and the control group, respectively. The weighted RR value can be represented by the following formula (Li et al., 2024):

$$\mathrm{RR{+}{+}} = \sum_i \left(RR_i \times W_i\right) \Big/ \sum_i W_i,$$

where i is the ith independent study. When the 95% CI of the value of RR++ includes 0, it indicates that the methane inhibitor treatment leads to a significant change in the effect size.

Given the variations in animals and experimental conditions among the studies, heterogeneity was anticipated; therefore, all analyses were performed using a random-effects model. The I^2 statistic was used to assess the magnitude of heterogeneity. The data collection workflow diagram was constructed by Reviewer Manager (RevMan) version 5.3 (The Cochrane Collaboration, London, UK). The sampling point diagram and principal component analysis diagram were created using an online website (https://www.chiplot.online/). Origin 2022 software (https://www.originlab.com) was used to create visual representations, including error charts, heat maps, chord diagrams, peak valley charts, principal component analysis diagrams, and regression curves.

RESULTS

Publication Bias Assessment

We generated funnel plots for each effect size and performed Egger's test to assess publication bias in the included results. Among all 94 metrics evaluated, the majority of funnel plots exhibited good symmetry; however, Egger's test results suggested potential publication bias in 21 metrics ($P < 0.05$), including: CP intake, NDF intake, BW, methane, microbial Chao1, archaeal β-diversity, bacterial structure, protozoan α-diversity, archaeal structure, *Butyrivibrio*, *Bacteroides*, Tenericutes, *Pseudobutyrivibrio*, *Selenomonas*, *Methanobrevibacter*, *Succinivibrio*, protozoa, methanogen, *Butyrivibrio fibrisolvens*, and *Butyrivibrio proteoclasticus*. To mitigate the effect of publication bias, we applied a Z-score filter to the results. Effect sizes with

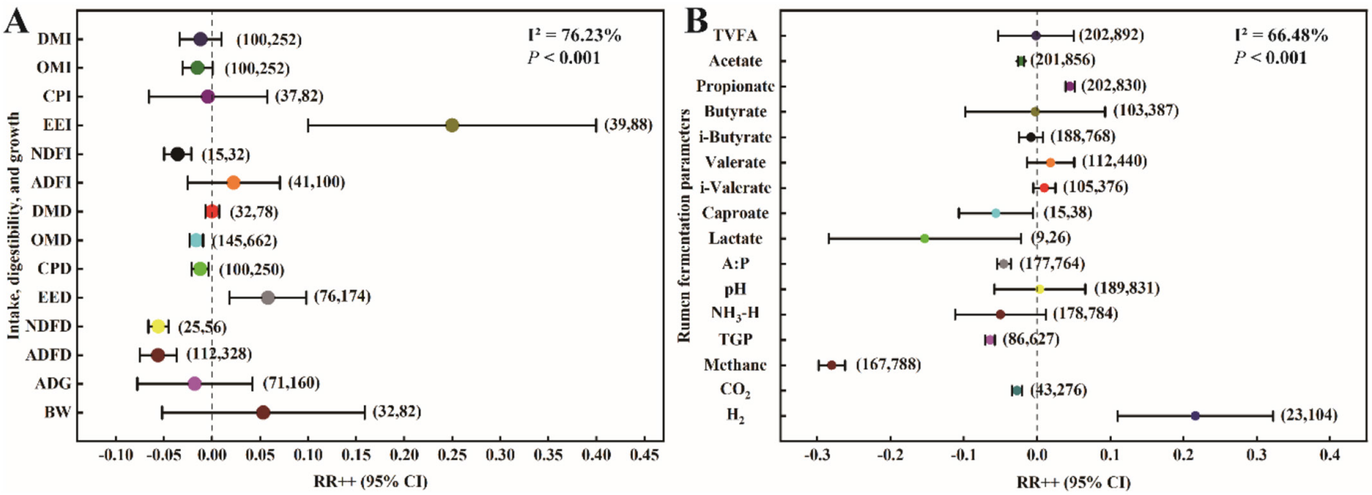

Figure 1. The effects of methane inhibitors on the performance and rumen fermentation of ruminants. (A) The effects of methane inhibitors on ruminant intake, digestion, and growth. (B) The effects of methane inhibitors on rumen fermentation in ruminants. The numbers in parentheses represent the number of articles and the number of samples. The I^2 represents the percentage of total variation between studies that is attributable to heterogeneity rather than sampling error. The RR++ represents the weighted effect size, with the lines around the mean indicating the 95% CI. The mean of RR++ value >0 indicates that the effect size of the experimental group is higher than that of the control group, whereas the mean <0 indicates that the effect size of the control group is higher than that of the experimental group.

an absolute Z-score greater than 3 were considered outliers and were excluded from the meta-analysis.

Effects on the Performance of Ruminants

Although the addition of methane inhibitors did not significantly reduce DMI and OM intake in ruminants ($P > 0.05$), the intake of feed was reduced. Similarly, the digestibility in ruminants generally decreased, especially with the digestibility of OM, CP, NDF, and ADF, which all significantly decreased ($P < 0.05$). In addition, the addition of methane inhibitors did not have a significant effect on BW ($P > 0.05$; Figure 1A, $I^2 = 76.23\%$, $P < 0.001$). In rumen fermentation, the addition of methane inhibitors significantly reduced the production of acetate, caproic acid, and lactic acid ($P < 0.05$). The ratio of acetate to propionate (**A:P**) was also significantly reduced ($P < 0.05$). In other effect sizes, the H_2 production significantly increased ($P < 0.05$), whereas the total gas production, methane, and carbon dioxide (CO_2) significantly decreased ($P < 0.05$). Additionally, hydrogen production and propionate levels significantly increased ($P < 0.05$; Figure 1B, $I^2 = 66.48\%$, $P < 0.001$).

Effects on the Rumen Microbiota

After adding methane inhibitors, the Shannon index of the rumen microbiota significantly decreased ($P < 0.05$), and the Chao1 index also decreased, although it was not significant ($P > 0.05$). In terms of microbial community structure, the addition of methane inhibitors did not significantly affect the overall microbial structure ($P > 0.05$). Subgroup analysis showed that the bacterial community's β-diversity was significantly increased ($P < 0.05$), whereas the β-diversity of archaea and protozoa was also significantly increased ($P < 0.05$; Figure 2A, $I^2 = 82.93\%$, $P < 0.001$). At the phylum level, *Firmicutes*, *Bacteroidetes*, *Proteobacteria*, *Spirochaetes*, *Tenericutes*, and *Verrucomicrobia* showed increased abundance, among which the increase in *Tenericutes* was significant ($P < 0.05$; Figure 2B, $I^2 = 88.57\%$, $P < 0.001$). At the genus level, the relative abundance of *Clostridium*, *Oscillospira*, *Prevotella*, *Succinivibrio*, and *Succiniclasticum* significantly increased ($P < 0.05$). The relative abundance of *Ruminococcus* significantly decreased ($P < 0.05$), whereas other bacterial genera showed no significant changes ($P > 0.05$). In archaea, except for a significant increase in the relative abundance of *Methanimicrococcus* ($P < 0.05$), no significant changes were observed in other genera ($P > 0.05$). At the genus level of protozoa, the relative abundance of *Isotricha* and *Ophryoscolex* significantly increased ($P < 0.05$), whereas *Diplodinium* and *Metadinium* significantly decreased ($P < 0.05$), and no significant changes were observed in other genera ($P > 0.05$; Figure 3C, $I^2 = 80.74\%$, $P < 0.001$). In the selected strains, the relative abundance of protozoa, methanogens, and fungi significantly decreased ($P < 0.05$). At the species level, the relative abundance of *Ruminococcus albus* and *Ruminococcus flavefaciens* significantly decreased ($P < 0.05$). The relative abundance of *Prevotella* spp., *Butyrivibrio proteoclasticus*, *Megasphaera elsdenii*, *Streptococcus bovis*, and *Butyrivibrio fibrisolvens* also significantly decreased ($P < 0.05$), whereas no significant changes were observed in other species ($P > 0.05$; Figure

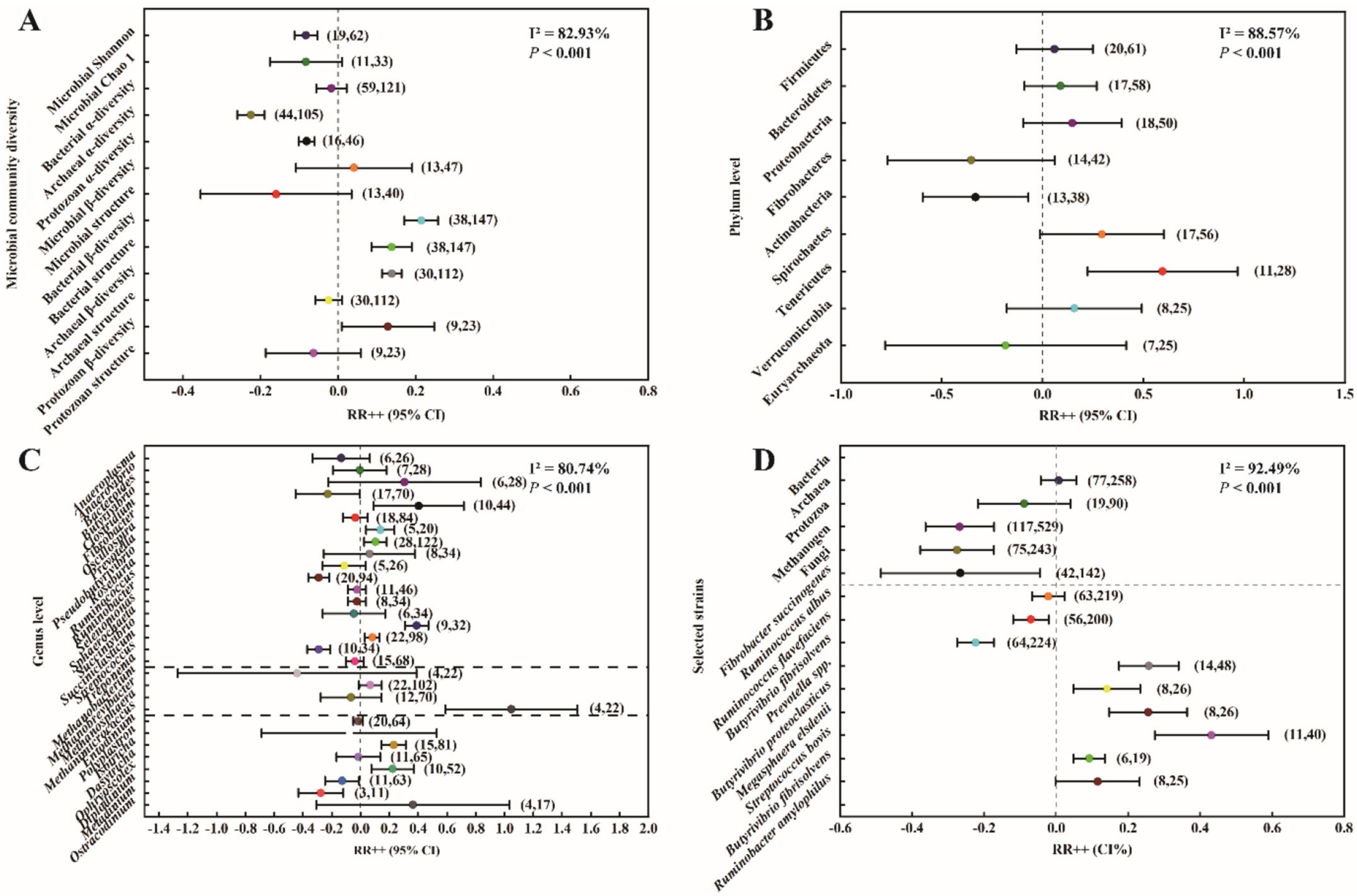

Figure 2. The effects of methane inhibitors on the rumen microbiota. (A) The effects of methane inhibitors on the α and β diversity of rumen microorganisms. (B) The effects of methane inhibitors on the relative abundance of rumen microorganisms at the phylum level. (C) The effects of methane inhibitors on the relative abundance of rumen microorganisms at the genus level. (D) The effects of methane inhibitors on the relative abundance of selected rumen microorganisms. The numbers in parentheses represent the number of articles and the number of samples. The I^2 represents the percentage of total variation between studies that is attributable to heterogeneity rather than sampling error. The RR++ represents the weighted effect size, with the lines around the mean indicating the 95% CI. The mean of RR++ value >0 indicates that the effect size of the experimental group is higher than that of the control group, whereas the mean <0 indicates that the effect size of the control group is higher than that of the experimental group.

2D, $I^2 = 92.49\%$, $P < 0.001$). The effect size distribution plot of methane inhibitors on the rumen microorganisms can be found in Supplemental Figure S6 (see Notes).

Effects of Different Types of Methane Inhibitors

The 4 principal coordinate analysis (**PCoA**) plots, respectively, show the effects of methane inhibitors on ruminant performance (Figure 3A, with principal coordinates 1 and 2 (**PCo1** and **PCo2**) accounting for 18.8% and 9.17%, respectively), on rumen fermentation in ruminants (Figure 3B, with PCo1 and PCo2 accounting for 9.29% and 6.33%, respectively), on the phylum-level composition of the rumen microbiota (Figure 3C, with PCo1 and PCo2 accounting for 28.64% and 21.97%, respectively), and on the genus-level composition of the rumen microbiota (Figure 3D, with PCo1 and PCo2 accounting for 20.66% and 15.68%, respectively). However, it is apparent from these plots that the effects of different methane inhibitor treatments were not clearly distinguished.

Effect of Methane Inhibitor Dose

The 4 PCoA plots, respectively, illustrate the effects of low-dose and high-dose methane inhibitors on ruminant performance (Figure 4A, with PCo1 and PCo2 accounting for 22.04% and 14.68%, respectively), rumen fermentation (Figure 4B, with PCo1 and PCo2 accounting for 29.83% and 9.42%, respectively), and the composition of rumen microbial communities at the phylum (Figure 4C, with PCo1 and PCo2 accounting for 62.43% and 19.27%, respectively) and genus levels (Figure 4D, with PCo1 and

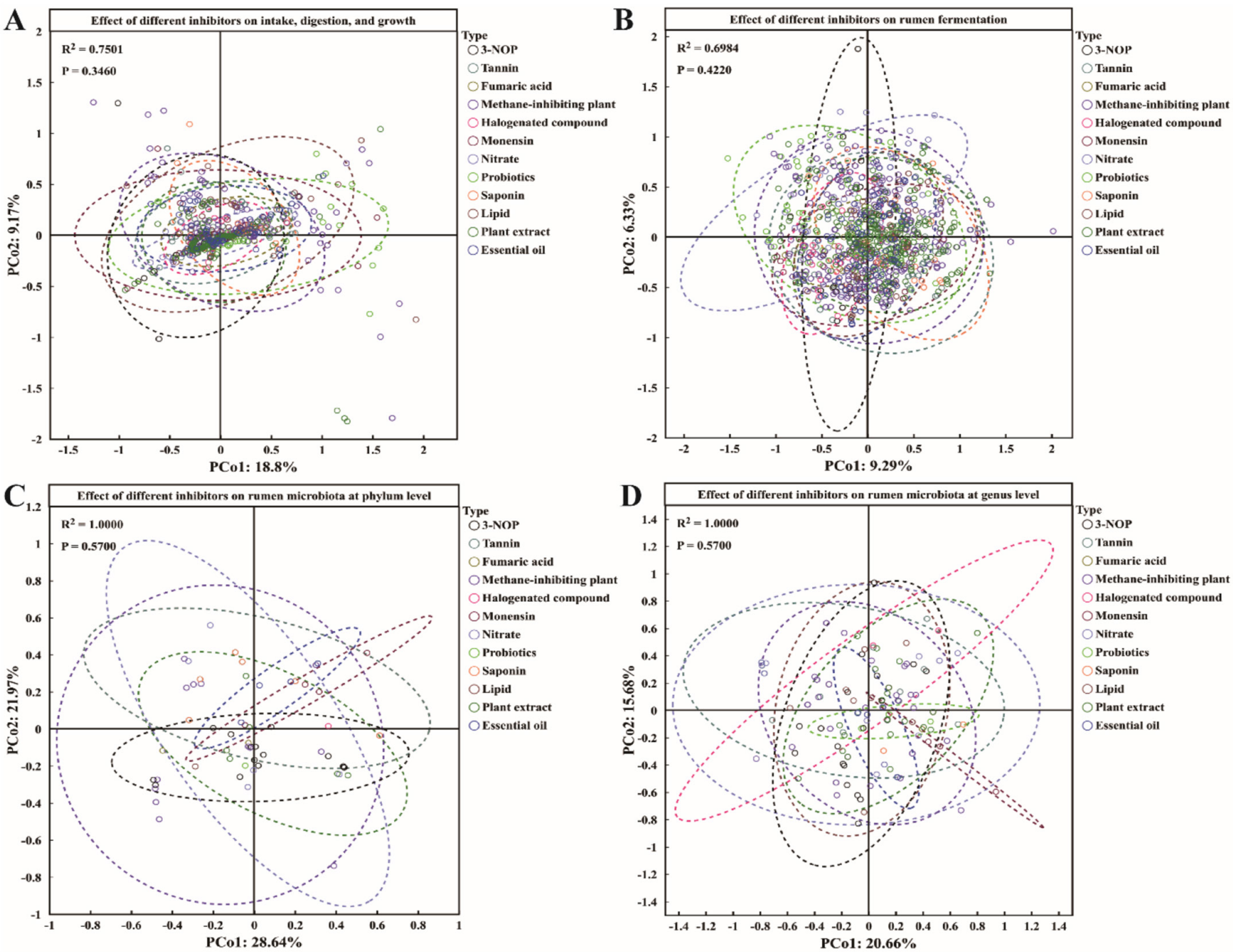

Figure 3. The differences in the effects of different methane inhibitors on ruminants. (A) The differences in the effects of different methane inhibitors on ruminant performance and rumen fermentation. (B) The differences in the effects of different methane inhibitors on rumen fermentation in ruminants. (C) The differences in the effects of different methane inhibitors on the phylum-level relative abundance of rumen microorganisms in ruminants. (D) The differences in the effects of different methane inhibitors on the genus-level relative abundance of rumen microorganisms in ruminants.

PCo2 accounting for 71.96% and 7.92%, respectively). The results showed that significant differences exist between the 2 treatment groups ($P < 0.05$), particularly in rumen fermentation and microbial community structure at both taxonomic levels.

Relationship Between Effect Sizes and Methane

The effect sizes of ruminant performance showed no significant correlation with methane yield ($P > 0.05$). The relationship between rumen fermentation effect sizes and methane yield showed significant positive correlations with total VFA (**TVFA**), acetate, and the A:P ($P < 0.05$), whereas correlations with other effect sizes were not significant ($P > 0.05$). Additionally, α-diversity of the rumen microbial community showed no significant correlation with methane yield ($P > 0.05$), and neither did the relative richness structure nor relative richness β-diversity ($P > 0.05$; Figure 5). The chord diagram between methane production and ruminant performance, methane inhibitors, and rumen fermentation parameters can be found in Supplemental Figure S7 (see Notes).

Linear Relationship Between Major Effect Sizes and Methane Emissions

Methane production gradually increases with the rise in TVFA, acetate, and the value of A:P. When fitting a

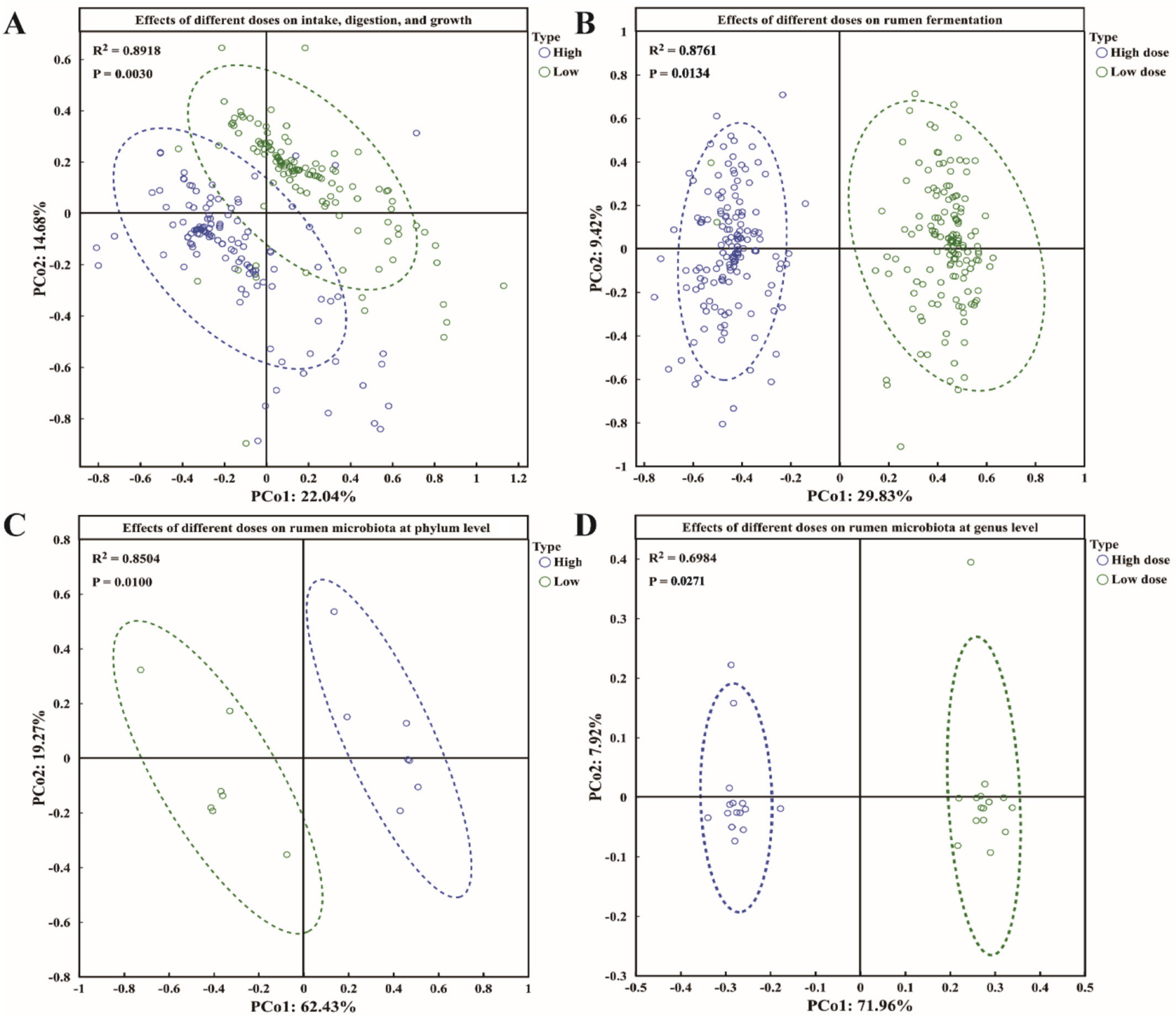

Figure 4. The differences in the effects of different doses of methane inhibitors on ruminants. (A) The differences in the effects of different doses of methane inhibitors on ruminant performance and rumen fermentation. (B) The differences in the effects of different doses of methane inhibitors on rumen fermentation in ruminants. (C) The differences in the effects of different doses of methane inhibitors on the phylum-level relative abundance of rumen microorganisms in ruminants. (D) The differences in the effects of different doses of methane inhibitors on the genus-level relative abundance of rumen microorganisms in ruminants.

quadratic polynomial model, the peak of the quadratic fitting curve between methane and acetate was 61.26 mmol/L ($R^2 = 0.48$, $P < 0.01$), the peak of the quadratic fitting curve between methane and TVFA was 96.98 mmol/L ($R^2 = 0.035$, $P < 0.01$), and the peak of the quadratic polynomial fitting curve between methane and the value of A:P was 3.86 ($R^2 = 0.021$, $P < 0.01$; Figure 6). The relationship between the effect size of methane production, the effect size of ruminant performance, the effect size of rumen fermentation parameters, and the effect size of rumen microbial α-diversity can be found in Supplemental Figures S3, S4, and S5 (see Notes).

DISCUSSION

Conducting systematic risk of bias assessment is an essential component during the literature screening process. The asymmetry observed in the funnel plots, coupled with significant Egger's test results, suggests potential under-publication of small-scale or statisti-

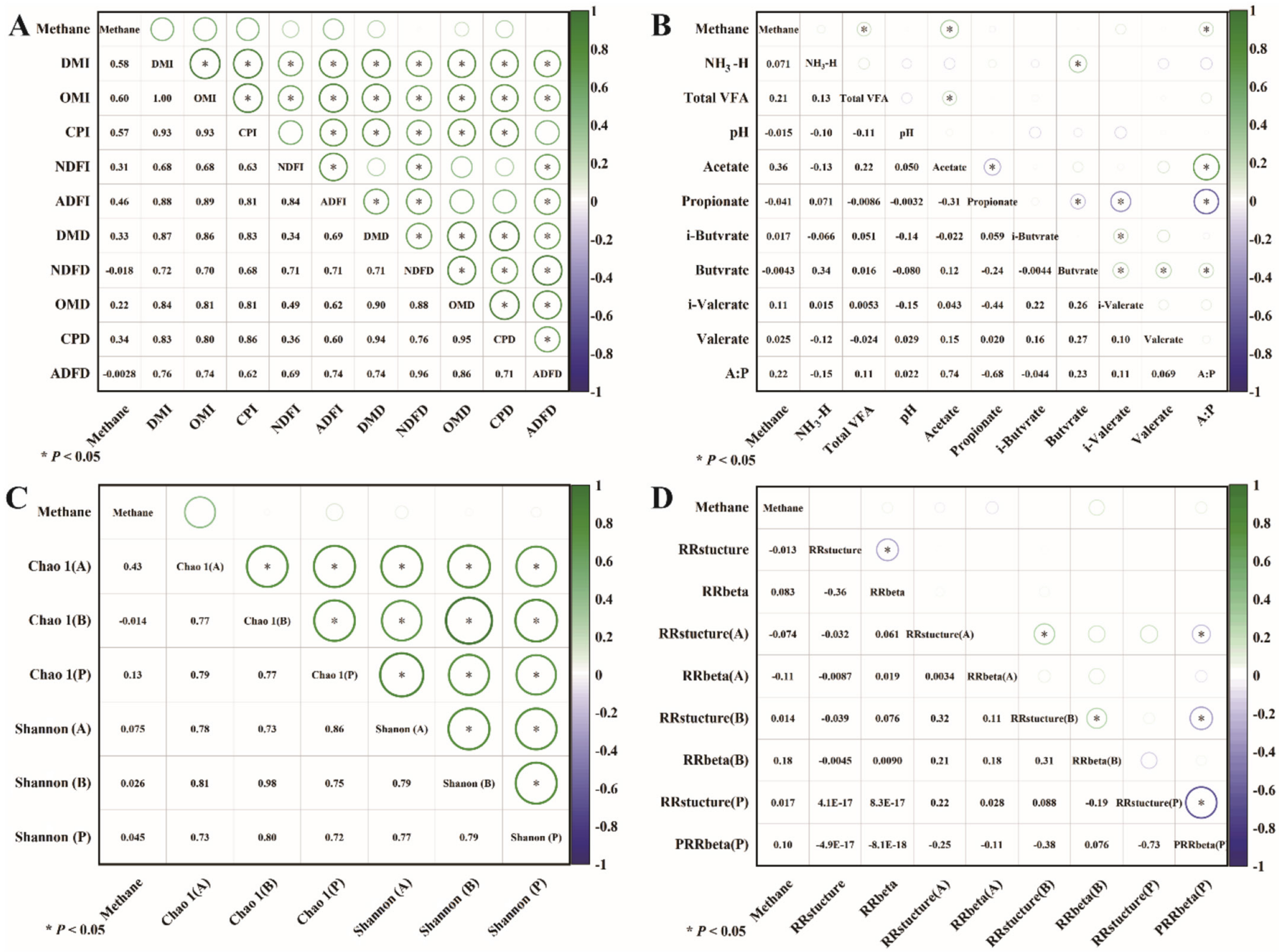

Figure 5. Heatmap of the correlation between methane production and various variables. (A) The relationship between methane production and ruminant performance. (B) The relationship between methane production and rumen fermentation in ruminants. (C) The relationship between methane inhibitors and rumen microbiota α-diversity. (D) The relationship between methane inhibitors and rumen microbiota β-diversity (* indicates significance at $P < 0.05$, A represents archaea, B represents bacteria, and P represents protozoa).

cally nonsignificant negative outcomes. This may lead to systematic overestimation of the overall effect size (Afonso et al., 2024). To evaluate the potential effect of this bias on the robustness of our conclusions, we performed a sensitivity analysis applying a Z-score criterion to identify and exclude extreme outliers. After removing outliers, the core effects remained stable and statistically significant. Furthermore, the preliminary conclusions of this study were not unduly influenced by individual extreme studies.

The reduction of methane not only helps control GHG emissions but also reduces energy losses in livestock (Kennedy and Charmley, 2012). It is important to note the effect of adding methane inhibitors on the intake, digestion, and growth performance of ruminants. Assuming that the addition of methane inhibitors has a positive effect on the feed intake, digestion, and growth of livestock, or that the reduction in methane production can offset its negative effects, it is considered that the methane inhibitors are beneficial to the performance of livestock. In fact, methane inhibitors reduce the feed intake of ruminants, but most studies suggest that their effect is not significant (Martinez-Fernandez et al., 2016; Araújo et al., 2023), and at the same time, the addition of methane inhibitors can lead to a decrease in the feed digestibility of livestock (Morgavi et al., 2023). The energy contained in the reduced methane cannot be simply redistributed and produced, which means that adding methane inhibitors does not improve animal performance (Ungerfeld, 2018). Our research results also support this, indicating that the energy from reduced methane emissions may not offset the negative effects of decreased feed intake and

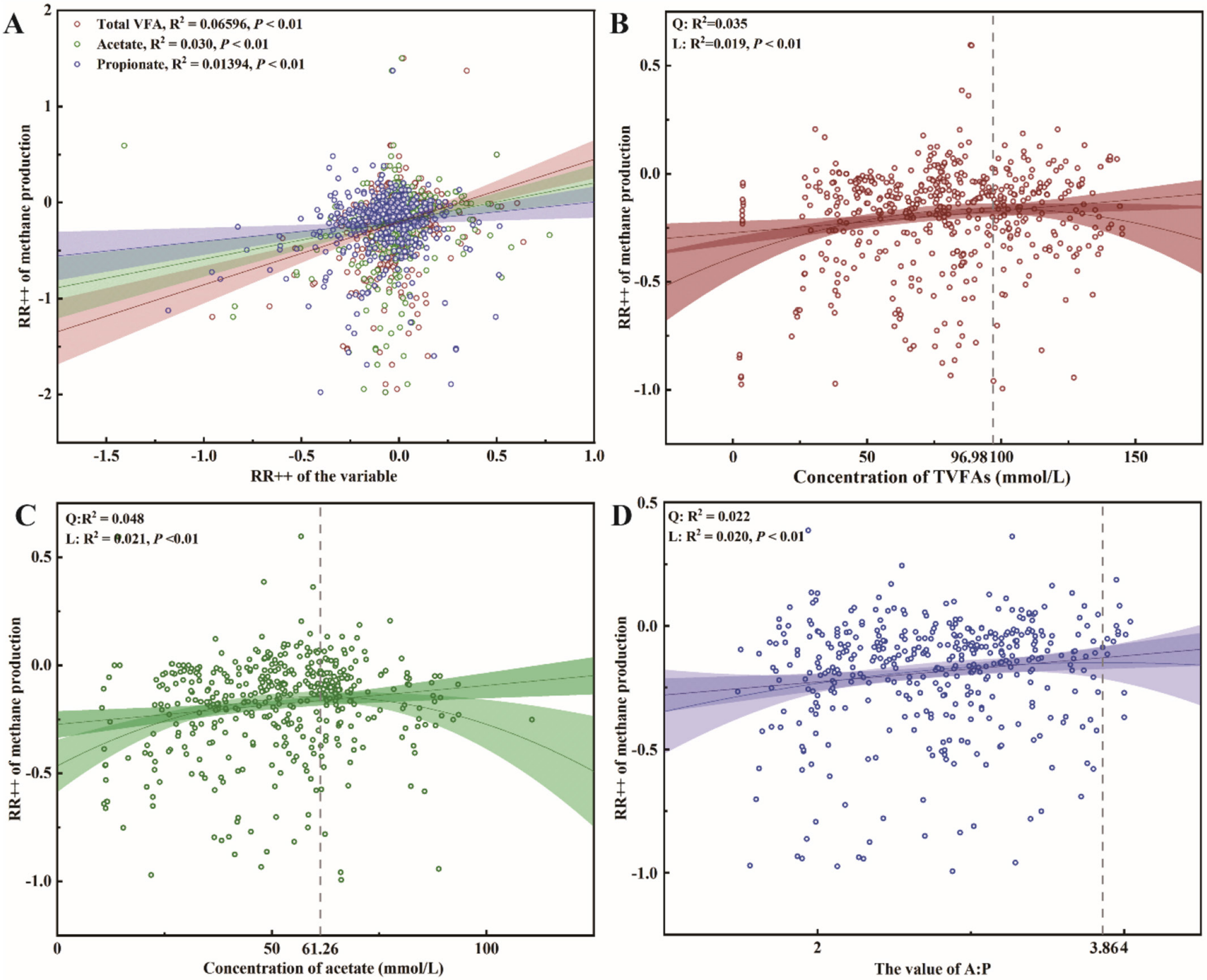

Figure 6. The linear relationship between methane production and TVFA, acetate acid, and A:P. (A) The trend relationship between methane production and TVFA, acetate, and A:P. (B) The linear relationship between methane production and TVFA. (C) The linear relationship between methane production and acetate. (D) The linear relationship between methane production and A:P. The R^2 represents the coefficient of determination, Q represents the quadratic curve, and L represents the linear curve. The dashed line represents the value on the x-axis at the vertex of the quadratic curve. The shaded area in the figure indicates the 95% CI.

digestibility, as the increase in BW is not statistically significant ($P > 0.05$). Previous studies have shown that the addition of methane inhibitors leads to the accumulation of H_2 in the rumen, and high concentrations of H_2 could inhibit microbial dehydrogenase activity, resulting in decreased feed intake and digestibility (Janssen, 2010). This is consistent with our research findings. The VFA in the rumen are also influenced by the partial pressure of hydrogen gas. When the partial pressure of hydrogen is high, it favors the production of propionate, although the opposite is true when the partial pressure of hydrogen is low, favoring the production of acetate (Roque et al., 2019; Glasson et al., 2022). The total VFA content is not affected by the addition of methane inhibitors (Morgavi et al., 2023). Relevant studies indicate that, compared with higher rumen pH values, lower rumen pH values lead to a reduction in methane emissions from dairy cows, with lower pH even resulting in a 54% decrease in methane emissions (Della Rosa et al., 2024).

For bacteria, researchers are more focused on the changes in fiber-degrading bacteria. For the rumen microbiota structure, the addition of methane inhibitors does not affect the total bacterial count but leads to significant changes in bacterial composition (Jayanegara et al., 2018; Kim et al., 2020). Due to the shift in acetate production to propionate production caused by methane inhibitors, it is estimated that the number of propionate-producing bacteria will increase (Ungerfeld and Pitta, 2025). This study observed an increase in the number of *Prevotella*, but the change in the number of *Selenomonas* was not significant. The potential reason is that the genus *Prevotella* encompasses multiple metabolically versatile species capable of utilizing hydrogen to promote propionate production (Betancur-Murillo et al., 2022), which may confer a competitive advantage under high-hydrogen conditions. In contrast, although certain *Selenomonas* species are also involved in propionate synthesis, their growth appears to rely more heavily on specific substrates such as lactate or succinate, rather than direct hydrogen utilization (Cabral and Weimer, 2024). Consequently, their response to changes in hydrogen partial pressure is less pronounced than that of *Prevotella*. The main fiber-degrading bacteria in the rumen include *Fibrobacter succinogenes*, *Ruminococcus flavefaciens*, and *Ruminococcus albus*. These bacteria may be inhibited due to the influence of hydrogen partial pressure. However, because *Fibrobacter succinogenes* does not produce hydrogen, it is not affected by methane inhibitors, whereas *Ruminococcus flavefaciens* and *Ruminococcus albus* may be inhibited due to the accumulation of hydrogen (Mitsumori et al., 2012). This is consistent with the results of this study. For archaea, suppressing methane production will reduce the abundance and composition of rumen archaea (Ungerfeld, 2018; Ungerfeld and Pitta, 2025). This study indeed observed a decrease in archaeal α-diversity and an increase in β-diversity. The number of *Methanobrevibacter*, among others, is not necessarily affected by methane inhibitors but is influenced by a combination of factors such as animal age, species, physiological state, and the type of methane inhibitor used (Martinez-Fernandez et al., 2017, 2018). The decrease or increase in protozoa abundance is not absolute (Dai and Faciola, 2019), and this may be related to the type of additives used. Protozoa provide the necessary hydrogen for methane production, and studies have shown that even in cases of hydrogen accumulation in the rumen, methane inhibitors do not lead to a decrease in protozoa numbers (Jayanegara et al., 2018; Kim et al., 2020); although other studies have reported that methane inhibitors can reduce the number of protozoa (Ando et al., 2003; Guyader et al., 2014).

Different methane inhibitors reduce methane emissions in different ways, and there are also differences in the effects of different doses of methane inhibitors on livestock. The different effects of various methane inhibitors may be due to their action on different mechanisms within the livestock gut microbiota. For example, 3-NOP, as a methyl-CoM analog, inhibits methane production and simultaneously reduces the relative abundance of *Methanobacteria* (Pitta et al., 2022b). Saponins can bind to sterol-like substances on the surface of protozoa, causing cell lysis (Goel et al., 2008), whereas essential oils not only inhibit the growth of protozoa but also increase the abundance of *Bacteroides* sp. and *Succinivibrio* sp. (Lei et al., 2019; Martins et al., 2024). Furthermore, the dose of these inhibitors affects the extent of microbial activity suppression. Higher doses may lead to more pronounced effects but could also result in potential side effects, such as changes in feed intake or overall animal health. In contrast, lower doses may be less effective at reducing methane emissions but carry fewer risks to the animals (Zhang et al., 2021). These findings suggest that optimizing the type and dose of methane inhibitors is crucial for achieving effective and sustainable methane reduction in livestock production.

The general view is that, without the addition of methane inhibitors, there is a significant correlation between methane emissions from ruminant animals and their feeding and digestion (McDonnell et al., 2016). In our study, no significant correlation was observed between methane production and feed intake and digestion. A possible reason for this could be that the addition of methane inhibitors hindered the influence of feed intake and digestion on methane production. However, the strong correlation between the concentration of VFA in the rumen and methane production may be used to simply predict methane output as an alternative to the current complex and cumbersome methane measurement methods (Niu et al., 2021). Although there is a highly significant positive correlation between the total VFA, acetate, propionate, and methane production in the rumen, this relationship does not follow a simple linear pattern (Williams et al., 2019). There is still no consensus on the specific concentrations of acetate and propionate in the rumen that would result in the maximum methane production efficiency. Although we used linear models, including both first-order and second-order curves, to fit methane production with TVFA, acetate, and propionate, and obtained the threshold values for the variables that lead to the highest methane production, the R^2 value was relatively small. The reason for this phenomenon is that the factors influencing methane production are not singular; they include genetic factors, diet quality, growth stages, environmental factors, and others (Wang et al., 2025). One molecule of glucose fermentation releases 8 molecules of [H] when yielding acetate, whereas 4 molecules of [H] are consumed when producing propionate. Although thermodynamically, lower concentrations of [H] in the rumen favor the production of acetate and methane, the formation of both acetate and methane in the rumen is in a dynamic equilibrium (Wang et al., 2023). Excessively high acetate concentrations lead to a reduction in [H] concentration, thereby inhibiting methane production. Although the fitted results in the study have limited biological significance, the variation does indeed follow a parabolic trend. Therefore, it is most effective to inhibit methane production only when the ratio of acetate concentration to propionate concentration in the rumen is below a certain level.

CONCLUSIONS

A comprehensive analysis of 256 articles collected for this study reveals that there are small differences in the overall effects of different methane inhibitors on ruminants. Moreover, the dosage of the inhibitor is more crucial than its type. The addition of methane inhibitors has a negative impact on the feed intake and digestion of ruminants, reduces the α-diversity of the rumen, but increases the β-diversity of the rumen, enhancing the stability of the overall microbial community structure in the rumen. Given the nonlinear relationship between methane emissions and the A:P, the threshold effect of VFA, and the potential inhibitory effects of methane inhibitors on feed intake and nutrient digestibility in ruminants, we recommend adopting a low-dose, continuous feeding strategy in practice to achieve an optimal balance between emission reduction benefits and production performance.

NOTES

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Nonstandard abbreviations used: 3-NOP = 3-nitrooxypropanol; A:P = ratio of acetate to propionate; PCoA = principal coordinate analysis; PCo1 and PCo2 = principal coordinates 1 and 2; TVFA = total VFA.

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